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Design:	Concrete - May 23, 2026	Date:	5/24/2026
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Specifier's comments:

1 Anchor Design

1.1 Input data

Anchor type and diameter:	HIT-HY 200 V3 + HAS-V-36 (ASTM F1554 Gr.36) 1/2	
Item number:	2198021 HAS-V-36 1/2"x4-1/2" (element) / 2334276 HIT-HY 200-R V3 (adhesive)	
Specification text:	HILTI Ø 1/2 IN HIT-HY 200 V3 + HAS-V-36 (ASTM F1554 GR.36) WITH 2.75 IN NOMINAL EMBEDMENT DEPTH PER ICC-ES ESR-4868 , HAMMER DRILL BIT INSTALLATION PER MPII	
Effective embedment depth:	$h_{ef,opti} = 2.750 \text{ in.}$ ($h_{ef,limit} = 10.000 \text{ in.}$)	
Material:	ASTM F1554 Grade 36	
Evaluation Service Report:	ESR-4868	
Issued Valid:	11/1/2024 11/1/2026	
Proof:	Design Method ACI 318-19 / Chem	
Shear edge breakout verification:	Row closest to edge (Case 3 only from ACI 318-19 Fig. R.17.7.2.1b)	
Stand-off installation:	$e_b = 0.000 \text{ in.}$ (no stand-off); $t = 0.500 \text{ in.}$	
Anchor plate ^{CBFEM} :	$l_x \times l_y \times t = 12.564 \text{ in.} \times 12.000 \text{ in.} \times 0.500 \text{ in.};$	
Profile:	Rectangular HSS (AISC), HSS4X2X.250; (L x W x T) = 4.000 in. x 2.000 in. x 0.250 in.	
Base material:	cracked concrete, 2500, $f'_c = 2,500 \text{ psi}$; $h = 420.000 \text{ in.}$, Temp. short/long: 32/32 °F	
Installation:	Hammer drilled hole, Installation condition: Dry	
Reinforcement:	tension: not present, shear: not present; no supplemental splitting reinforcement present edge reinforcement: none or < No. 4 bar	

^{CBFEM} - The anchor calculation is based on a component-based Finite Element Method (CBFEM)

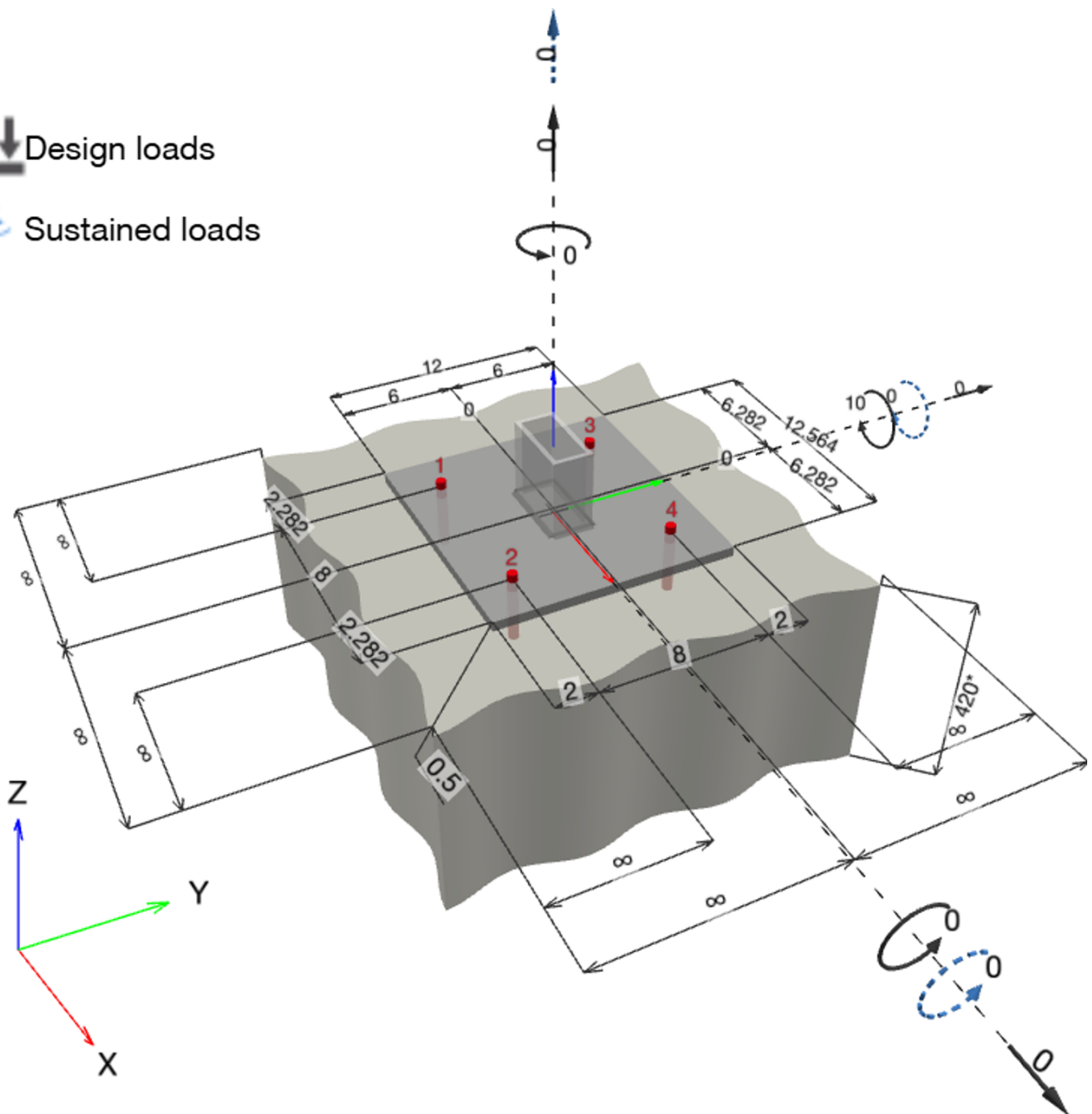
Geometry [in.] & Loading [kip, ft.kip]



Design loads



Sustained loads



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1.1.1 Design results

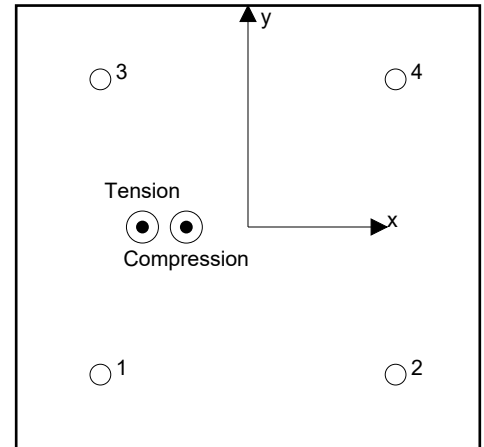
Case	Description	Forces [kip] / Moments [ft.kip]	Seismic	Max. Util. Anchor [%]
1	Combination 1	N = 0.000; $V_x = 0.000$; $V_y = 0.000$; $M_x = 0.00000$; $M_y = 10.00000$; $M_z = 0.00000$;	no	845

1.2 Load case/Resulting anchor forces

Anchor reactions [kip]

Tension force: (+Tension, -Compression)

Anchor	Tension force	Shear force	Shear force x	Shear force y
1	20.139	0.355	0.089	0.344
2	3.347	0.089	-0.089	0.008
3	20.130	0.355	0.089	-0.344
4	3.344	0.090	-0.089	-0.008



Resulting tension force in (x/y)=(-2.860/-0.001): 46.961 [kip]

Resulting compression force in (x/y)=(-1.671/-0.003): 60.900 [kip]

Anchor forces are calculated based on a component-based Finite Element Method (CBFEM)

1.3 Tension load

	Load N_{ua} [kip]	Capacity ϕN_n [kip]	Utilization $\beta_N = N_{ua}/\phi N_n$	Status
Steel Strength*	20.139	6.172	327	not recommended
Bond Strength**	46.961	5.561	845	not recommended
Sustained Tension Load Bond Strength*	N/A	N/A	N/A	N/A
Concrete Breakout Failure**	46.961	5.771	814	not recommended

* highest loaded anchor **anchor group (anchors in tension)



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1.3.1 Steel Strength

N_{sa} = ESR value refer to ICC-ES ESR-4868
 $\phi N_{sa} \geq N_{ua}$ ACI 318-19 Table 17.5.2

Variables

$A_{se,N}$ [in. ²]	f_{uta} [psi]
0.14	58,000

Calculations

N_{sa} [kip]
8.230

Results

N_{sa} [kip]	ϕ_{steel}	ϕN_{sa} [kip]	N_{ua} [kip]
8.230	0.750	6.172	20.139

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1.3.2 Bond Strength

$$N_{ag} = \left(\frac{A_{Na}}{A_{Na0}} \right) \psi_{ec1,Na} \psi_{ec2,Na} \psi_{ed,Na} \psi_{cp,Na} N_{ba} \quad \text{ACI 318-19 Eq. (17.6.5.1b)}$$

$$\phi N_{ag} \geq N_{ua} \quad \text{ACI 318-19 Table 17.5.2}$$

$$A_{Na} \text{ see ACI 318-19, Section 17.6.5.1, Fig. R 17.6.5.1(b)}$$

$$A_{Na0} = (2 c_{Na})^2 \quad \text{ACI 318-19 Eq. (17.6.5.1.2a)}$$

$$c_{Na} = 10 d_a \sqrt{\frac{\tau_{uncr}}{1100}} \quad \text{ACI 318-19 Eq. (17.6.5.1.2b)}$$

$$\psi_{ec,Na} = \left(\frac{1}{1 + \frac{e_N}{c_{Na}}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.5.3.1)}$$

$$\psi_{ed,Na} = 0.7 + 0.3 \left(\frac{c_{a,min}}{c_{Na}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.5.4.1b)}$$

$$\psi_{cp,Na} = \text{MAX} \left(\frac{c_{a,min}}{c_{ac}}, \frac{c_{Na}}{c_{ac}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.5.5.1b)}$$

$$N_{ba} = \lambda_a \cdot \tau_{k,c} \cdot \pi \cdot d_a \cdot h_{ef} \quad \text{ACI 318-19 Eq. (17.6.5.2.1)}$$

Variables

$\tau_{k,c,uncr}$ [psi]	d_a [in.]	h_{ef} [in.]	$c_{a,min}$ [in.]	$\alpha_{overhead}$	$\tau_{k,c}$ [psi]
2,220	0.500	2.750	∞	1.000	1,135
$e_{c1,N}$ [in.]	$e_{c2,N}$ [in.]	c_{ac} [in.]	λ_a		
2.860	0.001	3.988	1.000		

Calculations

c_{Na} [in.]	A_{Na} [in. ²]	A_{Na0} [in. ²]	$\psi_{ed,Na}$
7.071	490.27	200.00	1.000
$\psi_{ec1,Na}$	$\psi_{ec2,Na}$	$\psi_{cp,Na}$	N_{ba} [kip]
0.712	1.000	1.000	4.903

Results

N_{ag} [kip]	ϕ_{bond}	ϕN_{ag} [kip]	N_{ua} [kip]
8.556	0.650	5.561	46.961

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1.3.3 Concrete Breakout Failure

$$N_{cbg} = \left(\frac{A_{Nc}}{A_{Nc0}} \right) \psi_{ec,N} \psi_{ed,N} \psi_{c,N} \psi_{cp,N} N_b \quad \text{ACI 318-19 Eq. (17.6.2.1b)}$$

$$\phi N_{cbg} \geq N_{ua} \quad \text{ACI 318-19 Table 17.5.2}$$

$$A_{Nc} \text{ see ACI 318-19, Section 17.6.2.1, Fig. R 17.6.2.1(b)}$$

$$A_{Nc0} = 9 h_{ef}^2 \quad \text{ACI 318-19 Eq. (17.6.2.1.4)}$$

$$\psi_{ec,N} = \left(\frac{1}{1 + \frac{2 e_N}{3 h_{ef}}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.2.3.1)}$$

$$\psi_{ed,N} = 0.7 + 0.3 \left(\frac{c_{a,min}}{1.5 h_{ef}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.2.4.1b)}$$

$$\psi_{cp,N} = \text{MAX} \left(\frac{c_{a,min}}{c_{ac}}, \frac{1.5 h_{ef}}{c_{ac}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.2.6.1b)}$$

$$N_b = k_c \lambda_a \sqrt{f'_c} h_{ef}^{1.5} \quad \text{ACI 318-19 Eq. (17.6.2.2.1)}$$

Variables

h_{ef} [in.]	$e_{c1,N}$ [in.]	$e_{c2,N}$ [in.]	$c_{a,min}$ [in.]	$\psi_{c,N}$
2.750	2.860	0.001	∞	1.000
c_{ac} [in.]	k_c	λ_a	f'_c [psi]	
3.988	17	1.000	2,500	

Calculations

A_{Nc} [in. ²]	A_{Nc0} [in. ²]	$\psi_{ec1,N}$	$\psi_{ec2,N}$	$\psi_{ed,N}$	$\psi_{cp,N}$	N_b [kip]
264.06	68.06	0.591	1.000	1.000	1.000	3.876

Results

N_{cbg} [kip]	$\phi_{concrete}$	ϕN_{cbg} [kip]	N_{ua} [kip]
8.879	0.650	5.771	46.961



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1.4 Shear load

	Load V_{ua} [kip]	Capacity ϕV_n [kip]	Utilization $\beta_v = V_{ua} / \phi V_n$	Status
Steel Strength*	0.355	3.211	12	OK
Steel failure (with lever arm)*	N/A	N/A	N/A	N/A
Pryout Strength (Bond Strength controls)*	0.355	4.207	9	OK
Concrete edge failure in direction **	N/A	N/A	N/A	N/A

* highest loaded anchor ** anchor group (relevant anchors)

1.4.1 Steel Strength

V_{sa} = ESR value refer to ICC-ES ESR-4868
 $\phi V_{steel} \geq V_{ua}$ ACI 318-19 Table 17.5.2

Variables

$A_{se,V}$ [in. ²]	f_{uta} [psi]
0.14	58,000

Calculations

V_{sa} [kip]
4.940

Results

V_{sa} [kip]	ϕ_{steel}	ϕV_{sa} [kip]	V_{ua} [kip]
4.940	0.650	3.211	0.355

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1.4.2 Pryout Strength (Bond Strength controls)

$$V_{cp} = k_{cp} \left[\left(\frac{A_{Na}}{A_{Na0}} \right) \psi_{ed,Na} \psi_{cp,Na} N_{ba} \right] \quad \text{ACI 318-19 Eq. (17.7.3.1a)}$$

$$\phi V_{cp} \geq V_{ua} \quad \text{ACI 318-19 Table 17.5.2}$$

$$A_{Na} \text{ see ACI 318-19, Section 17.6.5.1, Fig. R 17.6.5.1(b)}$$

$$A_{Na0} = (2 c_{Na})^2 \quad \text{ACI 318-19 Eq. (17.6.5.1.2a)}$$

$$c_{Na} = 10 d_a \sqrt{\frac{\tau_{uncr}}{1100}} \quad \text{ACI 318-19 Eq. (17.6.5.1.2b)}$$

$$\psi_{ec,Na} = \left(\frac{1}{1 + \frac{e_N}{c_{Na}}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.5.3.1)}$$

$$\psi_{ed,Na} = 0.7 + 0.3 \left(\frac{c_{a,min}}{c_{Na}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.5.4.1b)}$$

$$\psi_{cp,Na} = \text{MAX} \left(\frac{c_{a,min}}{c_{ac}}, \frac{c_{Na}}{c_{ac}} \right) \leq 1.0 \quad \text{ACI 318-19 Eq. (17.6.5.5.1b)}$$

$$N_{ba} = \lambda_a \cdot \tau_{k,c} \cdot \pi \cdot d_a \cdot h_{ef} \quad \text{ACI 318-19 Eq. (17.6.5.2.1)}$$

Variables

k_{cp}	$\alpha_{overhead}$	$\tau_{k,c,uncr}$ [psi]	d_a [in.]	h_{ef} [in.]	$c_{a,min}$ [in.]	$\tau_{k,c}$ [psi]
2	1.000	2,220	0.500	2.750	∞	1,135
$e_{c1,N}$ [in.]	$e_{c2,N}$ [in.]	c_{ac} [in.]	λ_a			
0.000	0.000	3.988	1.000			

Calculations

c_{Na} [in.]	A_{Na} [in. ²]	A_{Na0} [in. ²]	$\psi_{ed,Na}$
7.071	122.57	200.00	1.000
$\psi_{ec1,Na}$	$\psi_{ec2,Na}$	$\psi_{cp,Na}$	N_{ba} [kip]
1.000	1.000	1.000	4.903

Results

V_{cp} [kip]	$\phi_{concrete}$	ϕV_{cp} [kip]	V_{ua} [kip]
6.009	0.700	4.207	0.355

1.5 Combined tension and shear loads, per ACI 318-19 section 17.8

β_N	β_V	ζ	Utilization β_{NV} [%]	Status
8.444	0.111	1.000	713	not recommended

$$\beta_{NV} = (\beta_N + \beta_V) / 1.2 \leq 1$$



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1.6 Warnings

- The anchor design methods in PROFIS Engineering require rigid anchor plates per current regulations (EN1992-4, ACI318, IS1946, AS 5216:2021, ETAG 001/Annex C, EOTA TR029 etc.). This means load re-distribution on the anchors due to elastic deformations of the anchor plate are not considered - the anchor plate is assumed to be sufficiently stiff, in order not to be deformed when subjected to the design loading. PROFIS Engineering calculates the minimum required anchor plate thickness with CBFEM to limit the stress of the anchor plate based on the assumptions explained above. The proof if the rigid base plate assumption is valid is not carried out by PROFIS Engineering. Input data and results must be checked for agreement with the existing conditions and for plausibility!
- The equations presented in this report are based on imperial units. When inputs are displayed in metric units, the user should be aware that the equations remain in their imperial format.
- Condition A applies where the potential concrete failure surfaces are crossed by supplementary reinforcement proportioned to tie the potential concrete failure prism into the structural member. Condition B applies where such supplementary reinforcement is not provided, or where pullout or pryout strength governs.
- Design Strengths of adhesive anchor systems are influenced by the cleaning method. Refer to the INSTRUCTIONS FOR USE given in the Evaluation Service Report for cleaning and installation instructions.
- For additional information about ACI 318 strength design provisions, please go to <https://viewer.joomag.com/profis-design-guide-us-en-summer-2021/0841849001625154758?short&/>
- Installation of Hilti adhesive anchor systems shall be performed by personnel trained to install Hilti adhesive anchors. Reference ACI 318-19, Section 26.7.
- The anchor design methods in PROFIS Engineering require rigid anchor plates per current regulations (EN1992-4, AS5216, etc.). This means load re-distribution on the anchors due to elastic deformations of the anchor plate are not considered - the anchor plate is assumed to be sufficiently stiff, in order not to be deformed when subjected to the design loading. PROFIS Engineering calculates the minimum required anchor plate thickness with FEM to limit the stress of the anchor plate based on the assumptions explained above. The proof if the rigid anchor plate assumption is valid is not carried out by PROFIS Engineering. Input data and results must be checked for agreement with the existing conditions and for plausibility!

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1.7 Installation data

Profile: Rectangular HSS (AISC), HSS4X2X.250; (L x W x T) = 4.000 in. x 2.000 in. x 0.250 in.

Hole diameter in the fixture: $d_f = 0.562$ in.

Plate thickness (input): 0.500 in.

Drilling method: Hammer drilled

Cleaning: Compressed air cleaning of the drilled hole according to instructions for use is required

Anchor type and diameter: HIT-HY 200 V3 + HAS-V-36 (ASTM F1554 Gr.36) 1/2

Item number: 2198021 HAS-V-36 1/2"x4-1/2" (element) / 2334276 HIT-HY 200-R V3 (adhesive)

Maximum installation torque: 0.03000 ft.kip

Hole diameter in the base material: 0.562 in.

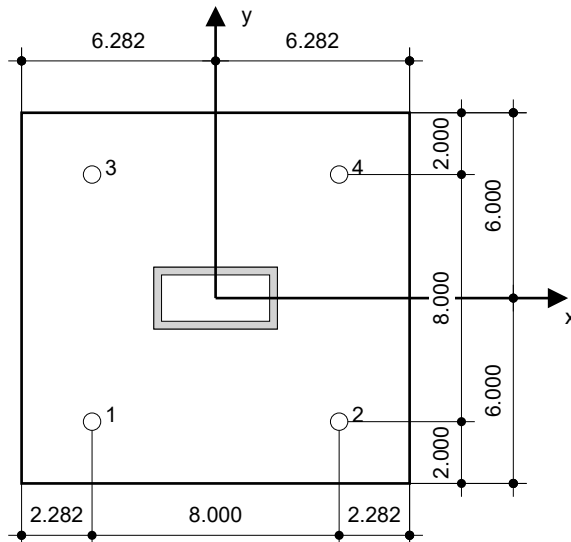
Hole depth in the base material: 2.750 in.

Minimum thickness of the base material: 4.000 in.

Hilti $\varnothing$ 1/2 in HIT-HY 200 V3 + HAS-V-36 (ASTM F1554 Gr.36) with 2.75 in nominal embedment depth per ICC-ES ESR-4868 , Hammer drill bit installation per MPII

1.7.1 Recommended accessories

Drilling	Cleaning	Setting
• Suitable Rotary Hammer • Properly sized drill bit	• Compressed air with required accessories to blow from the bottom of the hole • Proper diameter wire brush	• Dispenser including cassette and mixer • Torque wrench



Coordinates Anchor [in.]

Anchor	x	y	C _{-x}	C _{+x}	C _{-y}	C _{+y}
1	-4.000	-4.000	-	-	-	-
2	4.000	-4.000	-	-	-	-
3	-4.000	4.000	-	-	-	-
4	4.000	4.000	-	-	-	-



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2 Anchor plate design

2.1 Input data

Anchor plate:	Shape: Rectangular $I_x \times I_y \times t = 12.564 \text{ in} \times 12.000 \text{ in} \times 0.500 \text{ in}$ Calculation: CBFEM Material: ASTM A36; $F_y = 36,000 \text{ psi}$; $\epsilon_{lim} = 5.00\%$
Anchor type and size:	HIT-HY 200 V3 + HAS-V-36 (ASTM F1554 Gr.36) 1/2, $h_{ef} = 2.750 \text{ in}$
Anchor stiffness:	The anchor is modeled considering stiffness values determined from load displacement curves tested in an independent laboratory. Please note that no simple replacement of the anchor is possible as the anchor stiffness has a major impact on the load distribution results.
Design method:	AISC and LRFD-based design using component-based FEM
Stand-off installation:	$e_b = 0.000 \text{ in}$ (No stand-off); $t = 0.500 \text{ in}$
Profile:	HSS4X2X.250; ($L \times W \times T \times FT$) = $4.000 \text{ in} \times 2.000 \text{ in} \times 0.250 \text{ in} \times$ - Material: ASTM A500 Gr.B Rect; $F_y = 46,000 \text{ psi}$; $\epsilon_{lim} = 5.00\%$ Eccentricity x: 0.000 in Eccentricity y: 0.000 in
Base material:	Cracked concrete; 2500; $f_{c,cyl} = 2,500 \text{ psi}$; $h = 420.000 \text{ in}$
Welds (profile to anchor plate):	Type of redistribution: Plastic Material: E70xx
Mesh size:	Number of elements on edge: 8 Min. size of element: 0.394 in Max. size of element: 1.969 in

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2.2 Summary

	Description	Profile		Anchor plate		Hole bearing [%]	Welds [%]	Concrete [%]
		σ_{Ed} [psi]	ϵ_{Pl} [%]	σ_{Ed} [psi]	ϵ_{Pl} [%]			
1	Combination 1	49,162	10.89	44,259	28.45	2	290	86

2.3 Anchor plate classification

Results below are displayed for the decisive load combinations: Combination 1

Anchor tension forces	Equivalent rigid anchor plate (CBFEM)	Component-based Finite Element Method (CBFEM) anchor plate design
Anchor 1	6.599 kip	20.139 kip
Anchor 2	-0.001 kip	3.347 kip
Anchor 3	6.599 kip	20.130 kip
Anchor 4	-0.001 kip	3.344 kip

User accepted to consider the selected anchor plate as rigid by his/her engineering judgement. This means the anchor design guidelines can be applied.



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2.4 Profile/Stiffeners/Plate

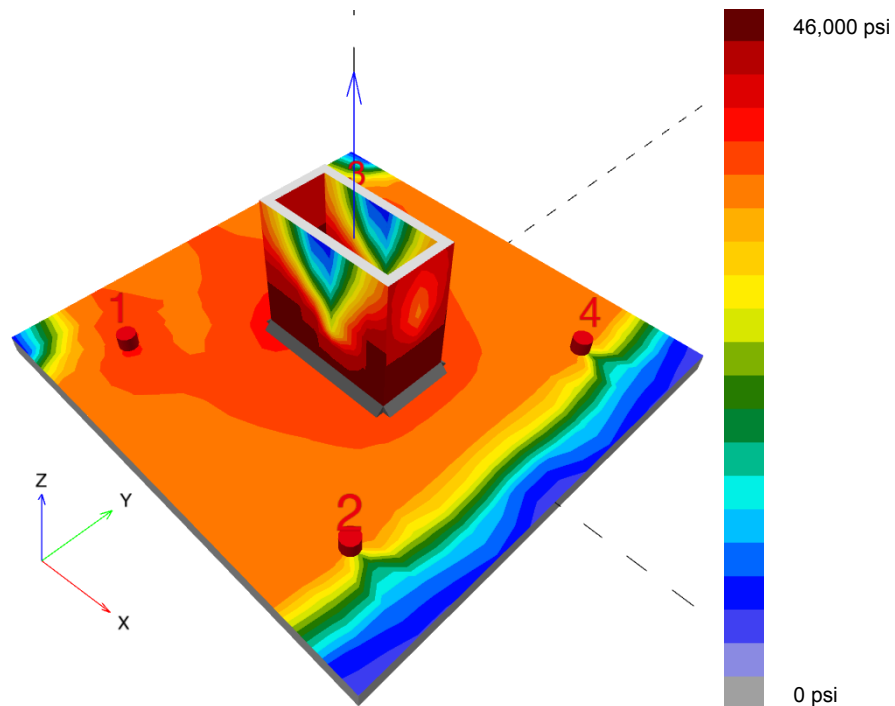
Profile and stiffeners are verified at the level of the steel to concrete connection. The connection design does not replace the steel design for critical cross sections, which should be performed outside of PROFIS Engineering.

2.4.1 Equivalent stress and plastic strain

Part	Load combination	Material	f_y [psi]	ϵ_{lim} [%]	σ_{Ed} [psi]	ϵ_{Pl} [%]	Status
Plate	Combination 1	ASTM A36	36,000	5.00	<u>44,259</u>	<u>28.45</u>	NOT OK
Profile	Combination 1	ASTM A500 Gr.B Rect	46,000	5.00	<u>48,825</u>	<u>9.73</u>	NOT OK
Profile	Combination 1	ASTM A500 Gr.B Rect	46,000	5.00	<u>49,162</u>	<u>10.89</u>	NOT OK
Profile	Combination 1	ASTM A500 Gr.B Rect	46,000	5.00	<u>48,219</u>	<u>7.64</u>	NOT OK
Profile	Combination 1	ASTM A500 Gr.B Rect	46,000	5.00	<u>48,228</u>	<u>7.67</u>	NOT OK

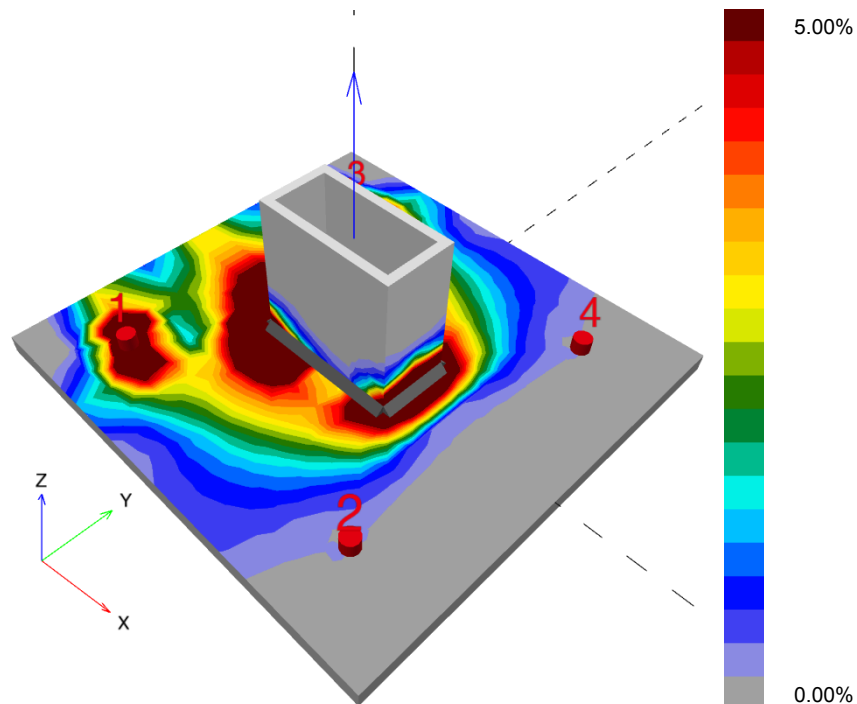
2.4.1.1 Equivalent stress

Results below are displayed for the decisive load combination: 1 - Combination 1



2.4.1.2 Plastic strain

Results below are displayed for the decisive load combination: 1 - Combination 1



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2.4.2 Plate hole bearing resistance, AISC 360-16 Section J3

Decisive load combination: 1 - Combination 1

Equations

$$R_n = \min(1.2 l_c t F_u, 2.4 d t F_u) \text{ (AISC 360-16 J3-6a, c)}$$

$$\phi R_n = 0.75 R_n$$

$$V \leq \phi R_n$$

Variables

	l_c [in]	t [in]	F_u [psi]	d [in]	R_n [kip]
Anchor 1	1.785	0.500	58,000	0.500	34.800
Anchor 2	2.010	0.500	58,000	0.500	34.800
Anchor 3	1.785	0.500	58,000	0.500	34.800
Anchor 4	2.010	0.500	58,000	0.500	34.800

Results

	V [kip]	ϕR_n [kip]	Utilization [%]	Status
Anchor 1	0.355	26.100	2	OK
Anchor 2	0.089	26.100	1	OK
Anchor 3	0.355	26.100	2	OK
Anchor 4	0.090	26.100	1	OK

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2.5 Welds

Profiles are modeled without taking the corner radius into account. Special rules for welding (e.g. for cold-formed profiles ...) are not taken into account by the software.

2.5.1 Anchor plate to profile

Decisive load combination: 1 - Combination 1

Equations

$$F_{nw} = 0.6 F_{EXX} (1.0 + 0.5 \sin^{1.5} \Theta)$$

$$\phi R_n = \phi F_{nw} A_w$$

$$\text{Utilization} = \frac{F_n}{\phi R_n}$$

Variables

Edge	X_u	T_h [in]	L_s [in]	L [in]	L_c [in]	F_{EXX} [psi]	Θ [°]	A_w [in ²]
Member 1-tfl 1	E70xx	0.157	0.223	1.992	0.498	70,000	66.5	0.08
Member 1-bfl 1	E70xx	0.157	0.223	1.992	0.498	70,000	90.0	0.08
Member 1-w 1	E70xx	0.158	0.223	3.486	0.498	70,000	88.5	0.08
Member 1-w 2	E70xx	0.158	0.223	3.486	0.498	70,000	88.5	0.08

Results

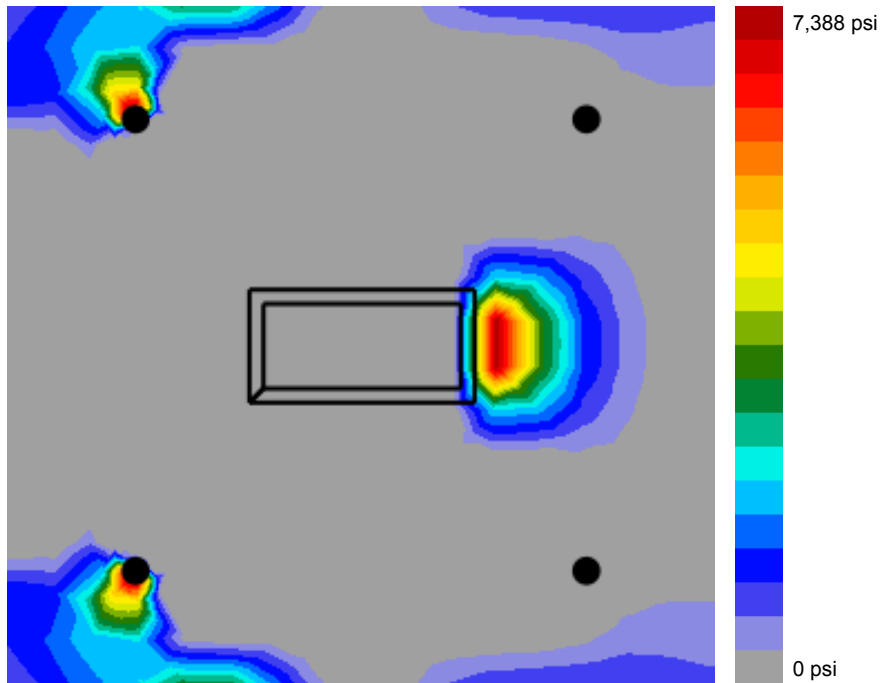
Edge	F_n [kip]	ϕR_n [kip]	Utilization [%]	Status
Member 1-tfl 1	10.306	3.556	290	NOT OK
Member 1-bfl 1	8.139	3.706	220	NOT OK
Member 1-w 1	6.300	3.710	170	NOT OK
Member 1-w 2	6.323	3.710	171	NOT OK

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2.6 Concrete

Decisive load combination: 1 - Combination 1

2.6.1 Compression in concrete under the anchor plate



2.6.2 Concrete block compressive strength resistance check, AISC 360-16 Section J8

Equations

$$F_p = \phi f_{p,max}$$

$$f_{p,max} = 0.85 f'_c \sqrt{\left(\frac{A_2}{A_1}\right)} \leq 1.7 f'_c; \sqrt{\left(\frac{A_2}{A_1}\right)} \leq 2$$

$$\sigma = \frac{N}{A_1}$$

$$\text{Utilization} = \frac{\sigma}{F_p}$$

Variables

N [kip]	f'_c [psi]	ϕ	A_1 [in ²]	A_2 [in ²]
60.900	2,500	0.65	25.92	573,810.14

Input data and results must be checked for conformity with the existing conditions and for plausibility!
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Results

Load combination	F _p [psi]	σ [psi]	Utilization [%]	Status
Combination 1	2,762	2,349	86	OK

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2.7 Symbol explanation

A_1	Loaded area of concrete
A_2	Supporting area
A_w	Effective area of weld critical element
d	Nominal diameter of the bolt
ε_{lim}	Limit plastic strain
ε_{Pl}	Plastic strain from CBFEM results
f_c	Concrete compressive strength
f'_c	Concrete compressive strength
F_{EXX}	Electrode classification number, i.e. minimum specified tensile strength
F_u	Specified minimum tensile strength of the connected material
F_n	Force in weld critical element
F_{nw}	Nominal stress of the weld material
F_p	Concrete block design bearing strength
$f_{p,max}$	Concrete block design bearing strength maximum
f_y	Yield strength
l_c	Clear distance, in the direction of the force, between the edge of the hole and the edge of the adjacent hole or edge of the material
L	Length of weld
L_c	Length of weld critical element
L_s	Leg size of weld
N	Resulting compression force
σ	Average stress in concrete
σ_{Ed}	Equivalent stress
ϕ	Resistance factor
ϕR_n	Factored resistance
R_n	Resistance
t	Thickness of the anchor plate
Θ	Angle of loading measured from the weld longitudinal axis
T_h	Throat thickness of weld
V	Resultant of shear forces V_y , V_z in bolt.
X_u	Filler metal tensile strength

2.8 Warnings

- By using the CBFEM calculation functionality of PROFIS Engineering you may act outside the applicable design codes and your specified anchor plate may not behave rigid. Please, validate the results with a professional designer and/or structural engineer to ensure suitability and



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- adequacy for your specific jurisdiction and project requirements.
- The anchor is modeled considering stiffness values determined from load displacement curves tested in an independent laboratory. Please note that no simple replacement of the anchor is possible as the anchor stiffness has a major impact on the load distribution results.



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3 Summary of results

Design of the anchor plate, anchors, welds and other elements are based on CBFEM (component based finite element method) and AISC.

	Load combination	Max. utilization	Status
Anchors	Combination 1	<u>845%</u>	NOT OK
Anchor plate	Combination 1	<u>123%</u>	NOT OK
Welds	Combination 1	<u>290%</u>	NOT OK
Concrete	Combination 1	86%	OK
Profile	Combination 1	<u>107%</u>	NOT OK

Fastening does not meet the design criteria!



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4 Remarks; Your Cooperation Duties

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